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Master Thesis

Evaluating the Effectiveness of Co-Teaching in Handling Noisy Labels in 3D Medical Imaging

by

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Evaluating the Effectiveness of Co-Teaching in Handling Noisy Labels in 3D Medical Imaging

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ABSTRACT

Context. Clinical diagnoses are often used as ground truth in medical datasets, but these labels can be incorrect, introducing label noise. Machine learning models are increasingly used in medical research to support early detection of Alzheimer's disease. However, models trained on noisy data risk learning from incorrect patterns, reducing diagnostic reliability. While handling label noise remains a challenge in the medical domain, extensive work in computer vision has led to the development of techniques designed to address noisy labels.

Goal. This study explores whether techniques developed in computer vision to handle noisy labels, specifically the Co-teaching method, can be effectively applied to medical datasets with diagnostic uncertainty.

Method. We build upon the Co-teaching method proposed by Han et al. [7] by applying it to two medical imaging datasets: OrganMNIST3D (CT-based organ classification) and ADNI (MRI-based Alzheimer's disease diagnosis). These datasets differ in both imaging modality and clinical complexity. To each dataset's train set, noise was injected: both unstructured (random symmetric flipping) and structured noise (pairflip or instance-dependent flipping) were introduced to reflect real-world error scenarios. Performance was evaluated using accuracy, balanced accuracy, and AUC, with confidence intervals generated via bootstrapping. *Results.* Our results show that Co-Teaching can significantly improve performance under both structured and unstructured label noise, particularly when class boundaries correspond to clear anatomical differences, as in organ classification. For example, on OrganMNIST3D with 50% symmetric noise, Co-teaching improved test accuracy from 64.2% (baseline) to 75.4%. In contrast, in Alzheimer's disease classification, overlapping features and diagnostic uncertainty made it harder to separate clean from noisy labels effectively. *Conclusions.* Co-teaching is effective for reducing the impact of label noise in medical image classification when class boundaries are clear and noise is synthetic. Its reliance on loss as a proxy for label correctness limits its utility in clinical datasets with high diagnostic uncertainty.

1 INTRODUCTION

Dementia affects more than 55 million people worldwide, with approximately 10 million new cases diagnosed each year [2]. This condition encompasses a range of illnesses, including Alzheimer's disease, vascular dementia, Lewy body dementia, and frontotemporal dementia. Among these, Alzheimer's disease is the most common, accounting for 60–70% of the cases. Dementia is now the seventh leading cause of death globally and one of the primary contributors to disability and dependency among the elderly [26]. Despite its prevalence, diagnosing dementia remains a significant challenge. According to Skinner et al. [21], diagnostic accuracy declines with age, leading to both over- and underdiagnosis. Overdiagnosis rates for Alzheimer's disease ranging from 16–40% often caused by overlapping symptoms with other conditions, especially in older adults. Overdiagnosis can lead to unnecessary treatment, emotional distress, and misinformed resource allocation [26]. Since pathological confirmation (e.g., via autopsy) is rarely available, clinical diagnoses are often used as ground truth in medical datasets. However, these labels can be incorrect, introducing label noise [20].

Machine learning models are increasingly used in medical research to support early detection of Alzheimer's disease and address the challenges associated with its diagnosis. These models are used because they are capable of identifying subtle patterns in imaging and clinical data, which can support a more accurate and earlier diagnosis, which is crucial in diseases like Alzheimer's. Although current medications cannot cure dementia or reverse its progression, they are most effective when administered in the early stages of the disease [1]. However, machine learning models trained on noisy data risk learning from incorrect patterns, which reduces diagnostic reliability, as these models often struggle with label uncertainty in domains with high diagnostic ambiguity. This complicates the development of reliable diagnostic tools [10]. While handling label noise remains a challenge in the medical domain, extensive work in computer vision, a branch of artificial intelligence (AI) that employs machine learning and neural networks to extract meaningful information from digital photos, has led to the development of techniques that are explicitly designed to address noisy labels. Understanding these methods provides valuable insights that could be applied to medical imaging tasks [11, 23, 30].

1.1 Research Aim

In this study, we aim to explore whether techniques developed in computer vision to handle noisy labels, specifically, the Co-teaching method, can be effectively applied to medical datasets with diagnostic uncertainty. The goal is to build upon the Co-teaching method proposed by Han et al. [7] by applying it to two medical imaging datasets: OrganMNIST3D, which contains CT scans for organ classification, and ADNI, which includes MRI scans for Alzheimer's disease diagnosis. These datasets differ in both imaging modality and clinical complexity. The goal is to evaluate Co-teaching's effectiveness in medical settings.Specifically, this study seeks to identify the conditions under which Co-teaching remains effective when training on medical data with noisy or uncertain labels, in order to determine when the method is most suitable for tackling diagnostic uncertainty in medical imaging. To contextualize its performance, we also compare Co-teaching's results on medical data with those

on natural image datasets such as CIFAR-10, where the method has been extensively validated.

The contributions of this study are as follows:

- Adaptation of the Co-teaching method to support 3D medical imaging inputs using the MONAI framework and Python 3.12.3.
- Design and implementation of two medical classification tasks using OrganMNIST3D (CT-based organ classification) and ADNI (MRI-based Alzheimer's disease diagnosis), with simulated label noise. Both unstructured noise (random symmetric flipping) and structured noise (structured pair flipping or instance-dependent flipping) were introduced to reflect different real-world error scenarios.
- Empirical evaluation of Co-teaching across multiple noise conditions and datasets, including comparisons with natural image datasets such as CIFAR-10.
- Analysis of loss distributions and model behaviour to understand the conditions under which Co-teaching is likely to be effective or fail in clinical applications.
- Derivation of practical guidelines to inform when Co-teaching is a suitable method for addressing label uncertainty in medical imaging tasks.

2 LITERATURE REVIEW

2.1 Label Noise in Deep Learning

Label noise refers to flaws, errors, or inconsistencies class labels in a training dataset for machine learning and artificial intelligence. These errors can be caused by a variety of factors, including human error during the labeling process, ambiguous data samples that are difficult to classify, and errors in automated labeling systems . In deep learning, such noise can mislead the training process and degrade model performance [17]. Therefore, learning with noisy labels is becoming a crucial task in medical image processing, since it reduces the generalization performance of deep neural networks. In the literature, several forms of label noise, such as symemtric , pairflip , and instance-dependent have been mentioned.

In symmetric noise the assumption is made that each label has a probability η of being flipped uniformly to any of the other $K-1$ classes, and a probability $1-\eta$ of remaining unchanged. This can be described using a noise transition matrix $T \in \mathbb{R}^{K\times K}$, where:

$$T_{i,j} = \begin{cases} 1-\eta & \text{if } i = j \\ \frac{\eta}{K-1} & \text{if } i \neq j \end{cases}$$

Wheras, Pairflip noise flips each label to one specific (predefined) alternative class with probability η, and leaves it unchanged with probability $1-\eta$. The transition matrix is defined as:

$$T_{i,j} = \begin{cases} 1-\eta & \text{if } j = i \\ \eta & \text{if } j = f(i) \text{ (the designated pair class)} \\ 0 & \text{otherwise} \end{cases}$$

Here, $f(i)$ maps class i to its assigned flip target, making this type of noise more structured and class-specific than symmetric noise. Moreover, instance-dependent noise occurs when a sample's corrupted label is determined by its true label and feature values in real-world situations (Liu, 2021; Zhang et al., 2021b) [13, 32]. In this study, we simulate both symmetric and structured (pairflip and instance-dependent) noise. The latter aims to simulate real-world challenges, such as overdiagnoses in older adults.

2.2 Approaches for Learning with Noisy Labels

2.2.1 Regularization-Based Methods. Due to their high capacity, deep neural networks are prone to overfitting, particularly when trained on noisy labels. One way to mitigate this issue is by introducing regularization terms to reduce the model's tendency to memorize noisy data. Regularization techniques usually have both explicit and implicit forms. Label smoothing (LS) is an example of an explicit form, while data augmentation and pre-training are examples of implicit forms. Label smoothing is a technique used to reduce the impact of label noise by softening the training labels. Instead of using hard labels (for instance [1,0,0]), LS replaces them with softer versions (for instance [0.9,0.05,0.05]) to reduce the model's overconfidence and improving robustness when some labels may be incorrect [20]. Lukasik et al. [14] applied this concept by mixing the original labels with a uniform distribution across all classes. However, Wei et al. [25] observed that standard LS becomes less effective in high-noise settings. They introduced Negative Label Smoothing (NLS), which applies different smoothing rates depending on the noise level, using stronger smoothing under high noise conditions and lighter smoothing when noise is low. Data augmentation might be useful when noisy labels are present, since this technique makes the training data more diverse. For example, Zhang et al. [31] created a method called MetaCleaner that adds variety to the data by generating clearer examples from noisy ones. Similarly, Nishi et al. [16] proposed a data augmentation strategy that combines both weak augmentations (e.g., horizontal flipping) and strong augmentations (e.g., AutoAugment) to enhance model robustness under label noise. Another promising approach involves pre-training. Hendrycks et al. [8] incorporated a pre-trained model into the Gold Loss Correction (GLC) framework and showed that while accuracy may remain stable, pre-training significantly improves the model's robustness against adversarial inputs. More recently, Zheltonozhskii et al. [33] proposed a self-supervised pre-training method to develop a robust feature extractor that is less sensitive to label errors. Implicit regularization methods have been shown to be successful in reducing overfitting and enhancing generalization, but they are not enough to address the problem of label noise (Song et al., 2022). Consequently, the community developed explicit regularization techniques like CDR (Xia et al., 2021) [27] and ELR (Early-Learning Regularization) (Liu et al., 2020) [13]. The foundation of ELR is the finding that there is an early-learning phase at the start of training during which the model learns the clean samples without overfitting the noisy ones. As a result, during the early learning stage, ELR steers the model output toward its own (correct) predictions by adding a regularization term to the Cross-Entropy loss. Likewise, CDR penalizes the non-critical parameters after first classifying the model parameters as critical or non-critical based on their significance for generalization.

2.2.2 Sample Selection-Based Methods. The core idea of sample selection based methods stems from research by Arpit et al. [3] where the memorization effects of deep neural networks were examined. They came to the conclusion that although deep networks

can learn to remember noisy labels, their findings indicate that they typically focus on learning basic patterns first. Neural networks that are trained on datasets with noisy labels initially match samples that are reasonably clean before gradually learning and overfitting samples that are quite noisy. This has led to a new approach for label-noise robust learning , where deep neural networks consider samples with low losses to be clean samples and use them to back-propagate new model parameters. One of the first methods built upon this approach was Decoupling (Malach & Shalev-Shwartz, 2017) [15], where two models are trained simultaneously but only update on samples where their predictions disagree, to prevent them from inheriting each other's errors. The MentorNet method, proposed by Jiang et al. [9] has a similar approach. MentorNet extends the sample selection idea by learning a data-driven curriculum to guide a main model (StudentNet) during training. Instead of relying on a fixed strategy, MentorNet adapts its training schedule based on feedback from StudentNet, deciding which samples deserve more attention at different stages of training. Once trained, MentorNet's parameters are frozen, and it is used to supervise StudentNet. Importantly, during inference, only StudentNet is used to make predictions. The Co-teaching method, proposed by Han et al. [7], builds upon this. Co-teaching is a deep learning paradigm that combats noisy labels by training two deep neural networks in parallel. In each mini-batch, each network selects a subset of low-loss (potentially clean) samples, shares them with its peer, and updates its parameters using only the samples selected by the other network [7].

2.2.3 The Hybrid Approach. In order to make fully use of all samples, researchers have proposed the hybrid approaches combining multiple orthogonal ideas. One of these approaches is DivideMix. DivideMix extends the Co-teaching idea using semi-supervised learning Specifically, it uses a Gaussian Mixture Model (GMM) to fit the per-sample loss distribution, which allows it to probabilistically divide training samples into two groups: a labeled set containing likely clean samples with low loss, and an unlabeled set containing potentially noisy samples with higher loss. Two networks are trained simultaneously, each using the clean/noisy split produced by the other, which prevents confirmation bias and reduces the risk of memorizing noisy labels [12]. Another hybrid approach that uses semi-supervised learning is Meta joint optimization (MJO) [19]. MJO is a hybrid framework that combines three complementary strategies for learning with noisy labels: (1) meta-reweighting based on a small, clean meta-set to estimate label quality; (2) semi-supervised consistency training on the relatively noisy subset; and (3) centroid-induced label updating, in which the labels of noisy samples are adjusted based on their feature distance to class centroids. In contrast to DivideMix , MJO does not rely on loss-based GMM partitioning and soft pseudo-labeling via MixMatch, but instead performs explicit label correction and only requires a small amount of clean data for metalearning. MJO distinguishes itself from DivideMix by simultaneously optimizing model parameters and label accuracy .

2.3 Application to Medical Imaging

In this study , the Co-teaching method was selected to test its effectiveness on medical datasets , due to its simplicity, mutual cross-updating mechanism, and proven effectiveness on benchmark datasets such as CIFAR-10 and MNIST [Han et al., 2018]. The Co-teaching method is compatible with many backbones and requires no prior knowledge of datasets, making it suitable for medical applications [34]. Especially compared to more complex hybrid approaches like DivideMix, which combines semi-supervised learning and requires careful tuning, Co-teaching offers a lightweight and interpretable solution that is better suited for practical use in medical settings. Furthermore, Co-teaching achieved superior performance compared to both regularization-based approaches like ELR and alternative sample selection strategies like MentorNet(Liu et al., 2020 ; Han et al., 2018) [7, 13]. Unlike ELR, which uses exponential moving averages of a single network's previous predictions, Co-teaching employs two distinct models, making it more resistant to confirmation bias and overfitting in high-noise environments. Moreover, MentorNet trains a separate network that learns to guide the training of the main model by assigning weights to samples based on their reliability, but it requires additional pretraining and lacks the mutual feedback mechanism of Co-teaching [7]. Additionally, we selected Co-teaching because it has already been successfully applied in medical imaging tasks, including skin lesion and thyroid ultrasound classification, where it achieved strong results [34]. However, there is still limited research on the application of noisy-label deep learning models, including Co-teaching, in dementia-related studies.

2.4 Problem Analysis

Although various strategies have been proposed to manage label noise in deep learning, their application within the medical field presents specific obstacles. Medical datasets tend to be relatively small, often display class imbalance, and are affected by structured forms of label noise that arise from diagnostic uncertainty rather than simple random mistakes [34]. For example, in the context of dementia diagnosis, clinical labels are frequently assigned without pathological confirmation, which can lead to consistent mislabeling, particularly in elderly individuals where symptoms may resemble those of other conditions [21]

Much of the research in label noise handling has focused on natural image datasets like CIFAR-10 and CIFAR-100. These datasets typically involve artificially added label noise that is evenly spread across all classes, a scenario that does not capture the structured and instance-dependent noise seen in medical contexts [20].

Moreover, while complex frameworks such as DivideMix and ELR have shown strong results, they may be less suited to clinical applications due to their computational demands and reliance on additional elements like clean validation data or prior assumptions about the noise type and limited, interpretability. Simpler techniques like co-teaching are easier to evaluate, adjust, and analyze, which makes them more useful in contexts where flexibility and transparency are crucial, like the healthcare industry. As such, there is a need for more straightforward and adaptable approaches that can address label uncertainty in healthcare data effectively.

This research explores the Co-teaching method using two real-world medical imaging datasets, OrganMNIST3D and ADNI (Alzheimer's Disease Neuroimaging Initiative), where both synthetic and structured noise are introduced. The objective is to determine how well

this approach performs under clinically relevant conditions and to identify its practical strengths and limitations.

3 METHODOLOGY

This study evaluates the Co-Teaching method on medical imaging data under various noise conditions. In Figure 4 an overview is shown of the experimental pipeline, from data preparation and noise injection to model training and evaluation. In the following subsections, each component is described in detail.

3.1 Co-Teaching

Co-teaching is a method designed to train deep networks reliably in the presence of extremely noisy labels, such as those used in fine-grained multi-class classification. The method employs two networks that simultaneously update their parameters. Each network selects instances with small losses (indicative of clean labels) and shares these instances with its peer for learning. This exchange of "cleaner" data samples helps to reduce the influence of noisy labeling.

The algorithm starts with two networks f and g, with parameters w_f and w_g. The dataset is divided into mini-batches $\bar{D}$, separately analyzed by each network. Each network calculates the loss value per data point; a smaller loss suggests a more likely clean label. Each network selects a portion of data points with the smallest losses ($\bar{D}_f$, $\bar{D}_g$), and shares them with the other:

$$L_f = \frac{1}{k} \sum_{i \in R_g} CE(f(x_i), t_i), \quad L_g = \frac{1}{k} \sum_{i \in R_f} CE(g(x_i), t_i) \tag{1}$$

Here, $CE(\cdot)$ denotes the cross-entropy loss function, x_i is the input sample, and t_i is the corresponding ground truth label. f and g are the two models trained simultaneously, and each uses a subset of samples selected by the other model.

R_f and R_g represent the indices of the small-loss samples selected by model f and g, respectively. Instead of training on all data, each model selects a percentage of samples in each mini-batch that it considers to be "clean" (i.e., with low loss), and passes those to its peer. This cross-update strategy helps reduce confirmation bias.

The value $k = (1 - R(T)) \cdot n$ denotes the number of selected (retained) samples per mini-batch, where $R(T)$ is the dynamic drop rate at epoch T. This drop rate increases linearly over time, allowing the model to gradually discard more suspected noisy labels as training progresses.

3.2 Experimental Design

3.2.1 Baseline Models. To evaluate the performance of the Co-Teaching method, it is compared against a baseline where a single convolutional neural network (CNN) is trained with standard cross-entropy loss without applying any noise-handling methods. This baseline model assumes that all labels in the dataset are correct (i.e., clean), even when noise is present. The goal of this baseline is to measure how much label noise affects performance when no correction or filtering is used. The baseline model is initialized as a 3D ResNet18 (from MONAI) model and uses Adam optimizer and the cross-entropy loss.

3.3 Adaptation to Clinical 3D Data

The original Co-Teaching framework by Han et al. (2018)[1] was implemented for 2D image classification using Python 2 and the CIFAR-10, CIFAR-100 and MNIST dataset. In this study , the code was adapted to support clinical 3D medical imaging data using Python 3.12.3 and the MONAI framework. To accommodate the volumetric input (e.g., `.nii.gz` and `.npy` files), a 3D version of ResNet-18 from the MONAI framework (`spatial_dims=3`) was integrated. Furthermore, all 3D images were resized to a fixed shape of `(96, 116, 96)` and normalized using MONAI's `NormalizeIntensity` transform. For OrganMNIST3D, additional augmentations were applied during training, including `RandRotate`, `RandGaussianNoise`, and `RandAdjustContrast`. The original loss function structure from Han et al. [7] was retained, which uses cross-entropy loss computed per sample, and applied the Co-Teaching strategy without modifications. For both ADNI and OrganMNIST3D, we logged per-sample loss and noise status, tracked the pure ratio per epoch (i.e., the percentage of clean samples among selected ones), and stored the discarded sample IDs. For OrganMNIST3D, models were trained for 200 epochs, following the original Co-Teaching implementation by Han et al.[7]. For ADNI, training was limited to 100 epochs, as visual inspection of the validation performance revealed signs of overfitting beyond this point. This decision is justified by the smaller sample size of the ADNI dataset compared to OrganMNIST3D and CIFAR-10. Models were trained with a batch size of 16 and evaluated using ensemble predictions from both models (mean logits).This setup facilitates a side-by-side evaluation of Co-Teaching under both synthetic 2D and realistic 3D clinical conditions, while maintaining consistent implementation across datasets.

3.4 Datasets

3.4.1 CIFAR-10. The CIFAR-10 dataset \cite{krizhevsky2009tiny} consists of 60,000 color images (size 32×32 pixels) categorized into 10 object classes (e.g., airplane, automobile, bird, cat, etc.). It includes 50,000 training and 10,000 test images with balanced class distributions. Each image was loaded as a PIL image and transformed using standard preprocessing (e.g., \texttt{ToTensor()}) The CIFAR-10 dataset is used in this study as a benchmark dataset. Since it was also employed in Han et al.[7], it provides a reference point to assess whether the Co-Teaching method can extend to medical imaging datasets.

3.4.2 OrganMNIST3D. The OrganMNIST3D dataset was sourced from the MedMNIST v2 collection (Yang et al., 2021) and consists of preprocessed abdominal CT scans labeled across 11 anatomical classes: liver, kidney-right, kidney-left, femur-right, femur-left, bladder, heart, lung-right, lung-left, spleen, and pancreas. Each 3D volume has been interpolated to a uniform resolution of $28 \times 28 \times 28$ and intensity-normalized to the range [0, 1]. No additional preprocessing was applied in this study. A total of 1,742 samples were included, which were split into a training set (1,393 samples; 79.97%) and a test set (349 samples; 20.03%) for use in the multi-class organ classification tasks. The distribution of samples, however, is not uniform across classes. For instance, in the training set , the

[1] https://github.com/bhanML/Co-teaching

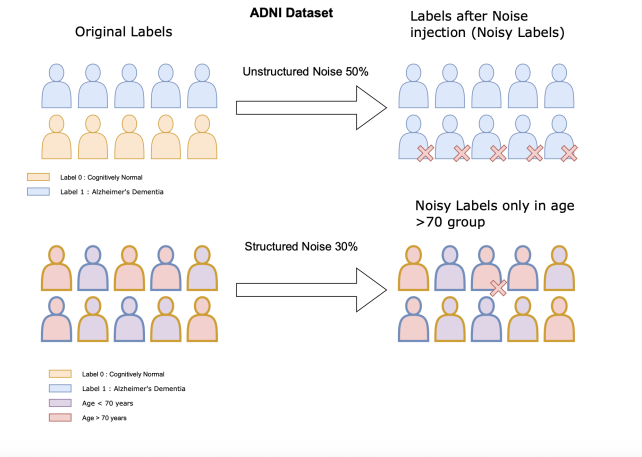


Figure 1: Illustration of unstructured (top) and structured (bottom) label noise on the ADNI dataset. Red X marks represent label flips due to noise. Structured noise is injected only in subjects older than 70 years.

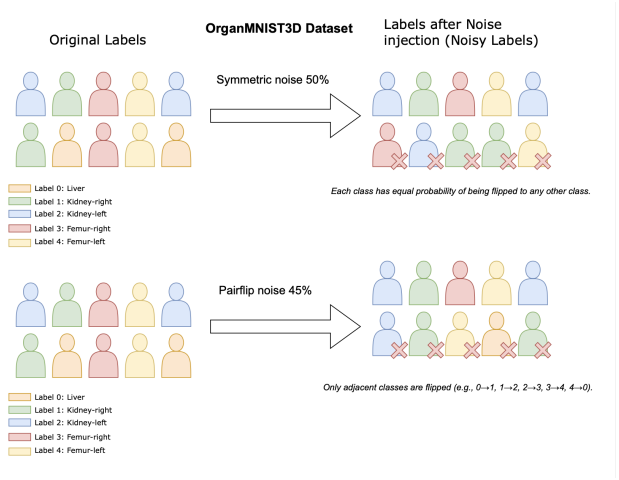


Figure 2: Illustration of symmetric (50%) and pairflip (45%) noise injection on the OrganMNIST3D dataset. Each class is represented with a unique color. In symmetric noise, labels are flipped randomly to any class; in pairflip noise, labels are flipped only to adjacent classes (e.g. class 0→1, 1→2, etc.). This figure is for illustrative purposes and does not reflect exact flipping proportions.

most frequent class contained 160 samples (11.49%), while the least frequent class contains only 58 samples (4.16%), resulting in a mild class imbalance ratio of 2.76×.

3.4.3 ADNI (Alzheimer's Disease Neuroimaging Initiative). Two binary classification tasks were performed using T1-weighted MRI brain scans [5]: Alzheimer's disease vs. cognitively normal. The total dataset comprised 746 samples (520 train, 226 test), with age ranges from 55.6 to 90.9 years. The ADNI MRI data were preprocessed using a voxel-based morphometry (VBM) pipeline [6, 18]. For gray matter segmentation, FreeSurfer v6.0 was used [4]. Nonlinear registration to the ICBM 152 template was performed to normalize anatomical variability across subjects. Spatial modulation via Jacobian determinants was applied to account for local volumetric changes. The resulting gray matter density maps have a standardized shape of $196 \times 232 \times 188$ voxels. Quality control procedures included automatic thresholding of outlier voxels and manual inspection to ensure anatomical plausibility. The ADNI train set used in this study contained 295 cognitively normal subjects (CN) and 225 subjects with Alzheimer's disease (AD), indicating a slightly unbalanced train set. To address the moderate imbalance of both the OrganMNIST3D dataset and the ADNI dataset, we report both Balanced Accuracy and AUC in all experiments to provide a fairer evaluation of the model's performance across classes.

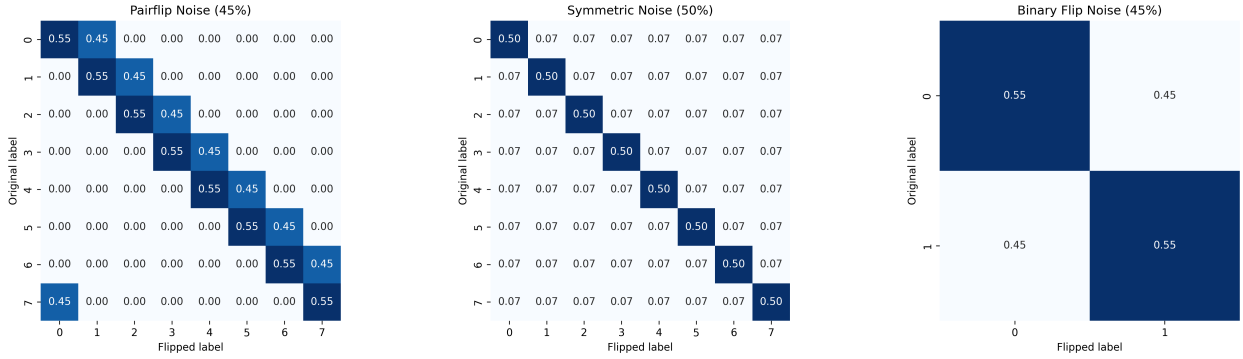
3.5 Noise Injection

To each dataset 's train set different noise injection methods were applied. In Figure 3 the transition matrices used to simulate the different label noise types are visualized. Each matrix represents the probability of a class label being flipped to another class during training, under different noise settings. This noise injection strategy was adapted from Han et al. [7] were two types of noise settings were introduced : Pairflip and Symmetric noise. In Pairflip noise (Figure 3a), when the label is affected by noise, it is forced to flip to the neighbouring class, rather than randomly to any other class. For example, class 0 flips to class 1 with a probability of 45%, while it remains correct with a probability of 55%. This flipping scheme is circular, meaning the last class flips back to the first (e.g., class 9 → class 0). Whereas in Symmetric noise , allows a label to be flipped to any other class with equal probability. As is visualized in Figure 3b, under 50% symmetric noise, each label has a 50% chance of remaining correct (diagonal value of 0.50) and a 50% chance of being flipped to one of the other classes. If there are 8 total classes, this 50% noise is equally distributed among the 7 incorrect classes, resulting in approximately 7.14% probability per incorrect label. This simulates a scenario where mislabeling is completely random and not class-specific).

3.5.1 CIFAR-10. In this study, we used a custom PyTorch implementation based on the original CIFAR-10 format ("cifar-10-batches-py") and introduced Symmetric noise and Pairflip noise. Noise rates of 20%, 50% (symmetric) and 45% (pairflip) were applied , following the experimental settings used in Han et al. [7]

3.5.2 OrganMNIST3D. Two types of noise were introduced: symmetric noise (random flipping) and pairflip noise. Noise rates of 20%, 50% (symmetric) and 45% (pairflip) were applied , following the experimental settings used in Han et al. [7]

3.5.3 ADNI. For the binary classification task in ADNI, traditional noise types such as symmetric or pairflip are no longer distinguishable. With only two classes, each label flip (regardless of direction or structure) has the same result: the label is changed to the lone alternative. Figure 3c shows a simple binary flip setup where each label has a fixed probability (e.g., 45%) of being flipped to the opposing class and a matching probability (e.g., 55%) of staying intact. As



(a) (b) (c)

Figure 3: ***Transition matrices for different noise types. (a) Pairflip noise with 45% noise rate, where each label is flipped only to its adjacent class. (b) Symmetric noise with 50% noise rate, where the label has an equal probability of being flipped to any other class. (c) Binary label noise example for a two-class setting (e.g., ADNI), where label flipping occurs between two classes. These visualizations clarify how different noise structures are applied to the training labels.***

a result, noise becomes intrinsically unstructured in binary settings, and the idea of class-specific flipping, which was previously applied in multi-class settings, is no longer applicable.

To still incorporate a form of structured noise , we used targeted label flipping from Alzheimer's Disease (AD) to Cognitively Normal (CN), limited to individuals aged 70 and up, which is more in line with clinical reality. This is similar to diagnostic ambiguity in older adults, when overlapping symptoms raise the likelihood of mislabeling. This process is visualized in Figure 2 Noise rates of 10%, 20%, 30%, 35%, 40%, and 45% were used to systematically evaluate the model's robustness under increasing levels of label corruption. This stepwise evaluation strategy deviates from the original Co-teaching paper by Han et al. [7], which focused on a limited number of fixed noise settings. Our design offers a clearer view of how performance changes with different noise levels, providing insights that are central to the purpose of this study.

3.6 Implementation Details

All experiments were conducted in Python 3.12.3 on Snellius, the Dutch national supercomputer.

4 RESULTS

The aim of this research is to apply the Co-teaching method proposed by Han et al. (2018) to medical datasets and evaluate its effectiveness in the medical domain. This was done using the OrganMNIST3D and ADNI datasets: the former contains abdominal CT scans labelled across 11 organ classes (e.g., liver, spleen, pancreas), forming a multi-class classification task. The latter consists of T1-weighted pre-processed brain MRI scans, divided into two classes: Alzheimer's Disease and Cognitively Normal (CN), forming a binary classification task. Both tasks were performed under different noise levels and noise types. In this section, we report the results of applying the Co-teaching method across various medical imaging experiments, with a focus on structured and unstructured label noise. We evaluate performance using three standard metrics: test accuracy, balanced accuracy, and area under the ROC curve (AUC). In addition, we report the pure ratio, as introduced by Han et al. (2018), to assess the proportion of clean samples among those selected for training. We also compare results against baseline models trained without noise handling to assess the added value of Co-teaching under different noise conditions and dataset characteristics.

4.1 Cifar-10: benchmark

To contextualize the results, the experiments were replicated on the CIFAR-10 dataset as in the original experiments by Han et al. [7]. Table 1 shows that the Co-teaching method outperformed the baseline in terms of test accuracy. Moreover, our test accuracy results were consistent with those presented in Han et al. (2018). In Han et al. (2018), a label precision metric was used to evaluate the effectiveness of sample selection in the presence of label noise. In each mini-batch, they selected a subset $R(T)$ of small-loss instances and computed the proportion of clean labels within that subset:

$$\text{Label Precision} = \frac{\text{number of clean labels}}{\text{number of all selected labels}}$$

A higher label precision indicates that fewer noisy instances are retained, suggesting better robustness to label noise. In this study, we refer to this label precision metric as the "pure ratio". Subsequently, the loss distributions of clean versus noisy samples were visualized in Figure 5 to assess their separability. These results show strong separability across epochs and noise settings, which aligns with the theoretical assumptions of Co-teaching.

Table 1: Final test accuracy for CIFAR-10 under different noise types

Noise Type	Baseline	Co-teaching
Pairflip 45%	51.52%	71.23%
Symmetric 50%	47.88%	74.50%
Symmetric 20%	74.99%	83.72%

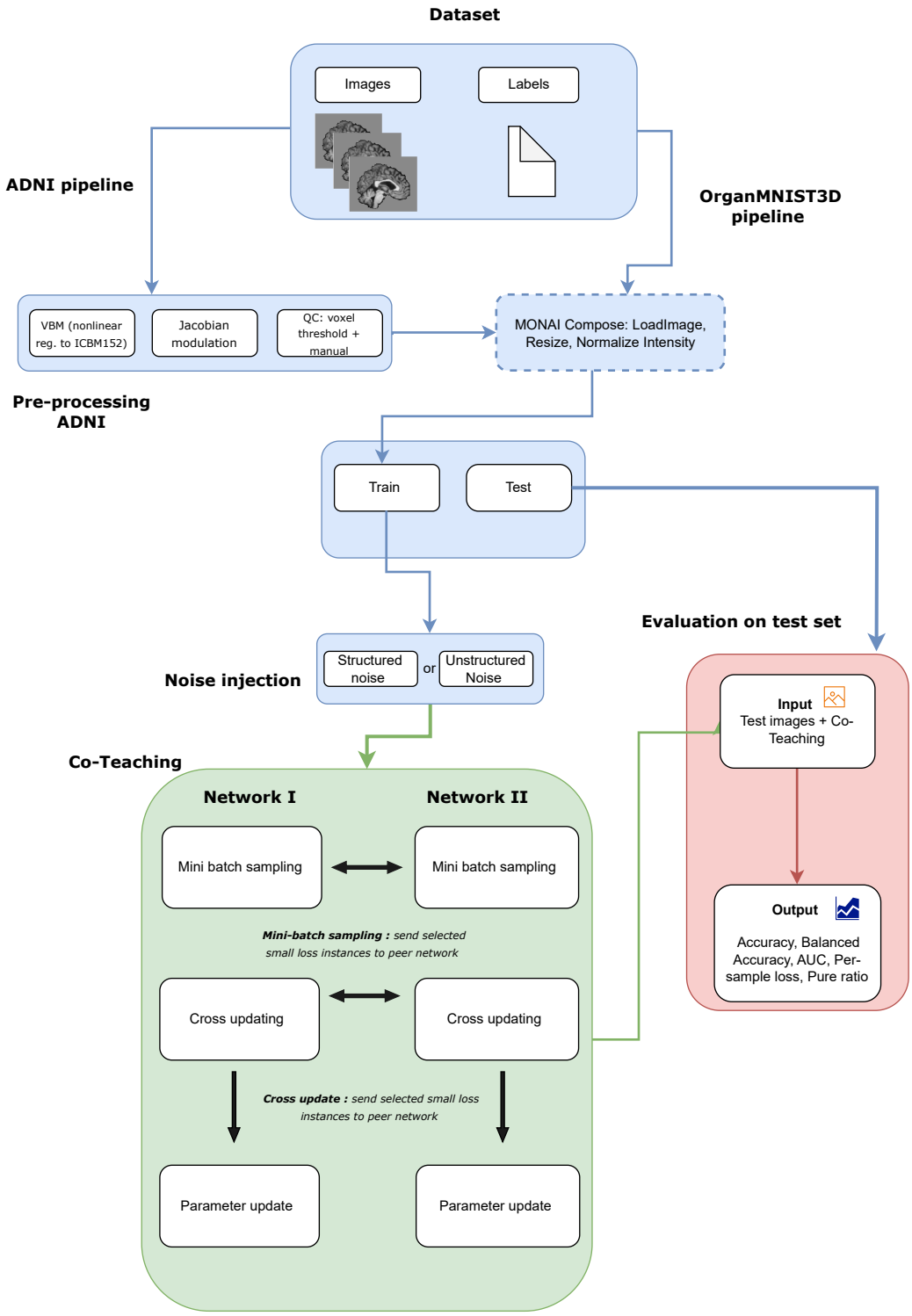
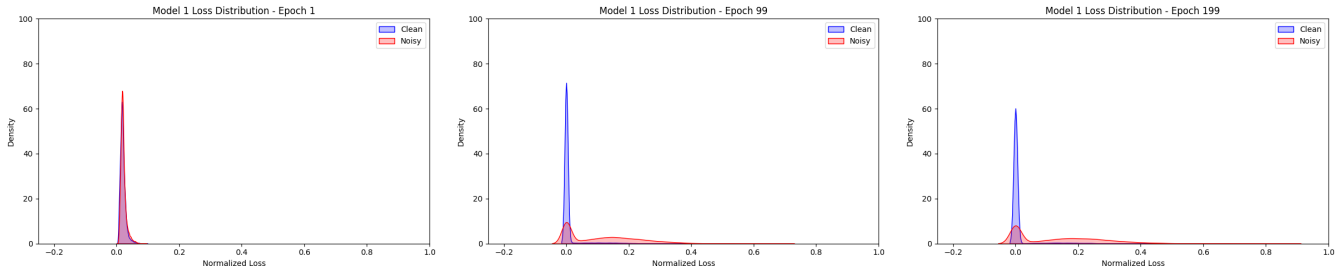


Figure 4: Overview of the Co-Teaching training and evaluation pipeline. ADNI and OrganMNIST3D follow distinct preprocessing paths. Label noise is injected before training, and evaluation is performed using accuracy, balanced accuracy, AUC, per-sample loss and pure ratio.



(a) (b) (c)

Figure 5: ***Loss distribution of Model 1 on CIFAR-10 with 45% symmetric noise at epochs 1, 99, and 199. Over time, loss separation becomes more distinct, aligning with Co-Teaching's assumption that clean samples have lower loss.***

4.2 OrganMNIST3D: multi-class classification

4.2.1 Test Accuracy Across Noise Conditions. Table 2 reports the performance of Co-teaching under three noise conditions: symmetric noise (20%, 50%) and pairflip noise (45%). At 50% symmetric noise, Co-teaching achieved 75.36% vs. 64.18% (baseline); at 45% pairflip noise, 57.31% vs. 52.72%. Balanced accuracy was also included in Table 2 to account for class imbalance in OrganMNIST3D. It followed a similar trend, further confirming the observed results across all organ classes. Moreover, Table 2 also reports the confidence intervals of all the metrics. For the final accuracy , only the symmetric 50% noise condition exhibited non-overlapping confidence intervals, indicating a statistically significant difference.

4.2.2 AUC Performance. Table 2 shows that Co-teaching achieved a higher macro AUC across symmetric noise settings, but fails to outperform the baseline model on the 45% pairflip noise setting. The confidence intervals indicate that this result is significant, since the intervals do not overlap. Moreover, the symmetric 50% condition also exhibited non-overlapping confidence intervals, indicating statistical significance, whereas no significant difference was observed in the symmetric 20% condition.

4.2.3 Loss Distributions of Clean vs. Noisy Samples. Figure 6 shows the loss distributions of clean versus noisy samples at early, mid, and late epochs under the 45% pairflip noise condition. In contrast to the 20% and 50% symmetric noise settings, where clean and noisy samples were relatively well-separated (see Supplementary Figures 11 and 12), the pairflip setting exhibited substantial overlap between the two distributions throughout training. This likely contributed to the model's reduced ability to filter mislabeled samples effectively and may explain the drop in performance under this condition. Furthermore, as shown in Supplementary Figure **??**, the average training loss under 45% pairflip noise fluctuates and remains consistently higher than in the symmetric noise settings. This instability further supports the observation that Co-teaching struggles under pairflip noise on the OrganMNIST3D dataset.

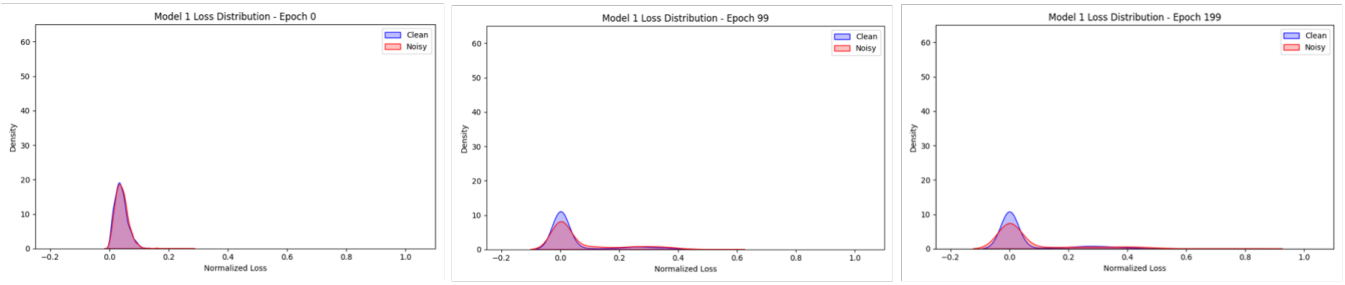
4.2.4 Comparison to CIFAR-10 Results. Although absolute accuracy values are not directly comparable between CIFAR-10 and OrganMNIST3D due to differences in domain and task complexity, the relative effect of the noise types within each dataset was analyzed to explore the potential extension of this method to medical imaging. The test accuracy of Co-teaching significantly drops when dealing with pairflip noise on the OrganMNIST3D dataset, especially when compared to CIFAR-10 results reported in Han et al. (2018). On CIFAR-10, the difference between symmetric noise (50%) and pairflip noise (45%) was small, Co-teaching achieved 75.53% vs. 72.62%, a drop of about 3%. On OrganMNIST3D, this difference was much larger: 75.36% vs. 57.31%, a drop of over 18%. This suggests that pairflip noise is more challenging for medical data than for natural images. Moreover, on CIFAR-10, the loss distributions of clean and noisy samples gradually diverge over time, allowing Co-teaching to distinguish and filter noisy labels more effectively. In contrast, the OrganMNIST3D loss distributions show substantial overlap throughout the training. Even at later epochs (e.g., epoch 199), the model struggles to separate noisy from clean samples. This is further supported by the observed pure ratio trends. In Supplementary Figure 33, the pure ratios under pairflip noise remain consistently lower for OrganMNIST3D compared to the symmetric noise settings. This suggests that the models had more difficulty identifying and retaining clean samples in the pairflip setting.

4.3 ADNI: Unstructured Noise

4.3.1 Test Accuracy Across Noise Conditions. Co-teaching was applied to the ADNI dataset under unstructured label noise with flipping rates of 20%, 30%, 35%, and 45%. Co-teaching consistently outperformed the baseline in all noise settings, with the largest gain observed at 20% noise: 77.43% vs. 61.50 (Table 3). However, the performance gain decreased as the noise increased. At the highest noise level (45%), Co-teaching still performed better (56.64%) than the baseline (43.36%), but with a smaller margin. However, the test accuracy drops significantly when the noise level increases beyond 20%. Moreover, in Table 3, balanced Accuracy was also included to account for the slight class imbalance in ADNI. Similar to the OrganMNIST3D results, the differences between test accuracy and balanced Accuracy remained small (all <1%), indicating that performance was not heavily biased towards the majority class. The confidence intervals displayed in Table 3 confirm the statistical significance of Co-teaching's improvement across all metrics at low noise (20%) and high noise (45%), but show overlapping Confidence intervals at noise levels of 30% and 35%.

Table 2: Final test accuracy, balanced accuracy (in %), and macro AUC for OrganMNIST3D under different noise types. Confidence intervals (95%) are reported next to the corresponding metric values.

Noise Type	Baseline			Co-teaching		
	Acc (%)	Balanced Acc (%)	Macro AUC	Acc (%)	Balanced Acc (%)	Macro AUC
Symmetric 20%	89.7 [86.0–92.3]	90.6 [87.3–93.2]	0.99 [0.99–1.00]	90.0 [86.0–92.6]	90.6 [87.0–93.2]	1.00 [0.99–1.00]
Symmetric 50%	64.2 [58.7–68.8]	63.4 [57.7–68.6]	0.93 [0.911–0.947]	75.4 [70.5–79.4]	75.9 [70.7–80.3]	0.98 [0.97–0.98]
Pairflip 45%	52.7 [47.3–57.6]	52.9 [47.2–58.6]	0.93 [0.921–0.943]	57.3 [51.9–62.5]	58.1 [54.4–61.6]	0.85 [0.83–0.86]



(a) (b) (c)

Figure 6: ***Loss distribution of Model 1 on OrganMNIST3D with 45% pairflip noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 99, 199). Over time, the clean and noisy samples become increasingly inseparable in their loss values, which is inconsistent with the assumption that clean samples converge to lower loss.***

4.3.2 AUC performance. Table 3 shows that Co-teaching achieved a notably higher AUC at 20% noise, but the differences decrease with increasing noise levels. This suggests that Co-teaching is beneficial under moderate noise conditions on the ADNI dataset if the label noise is random.

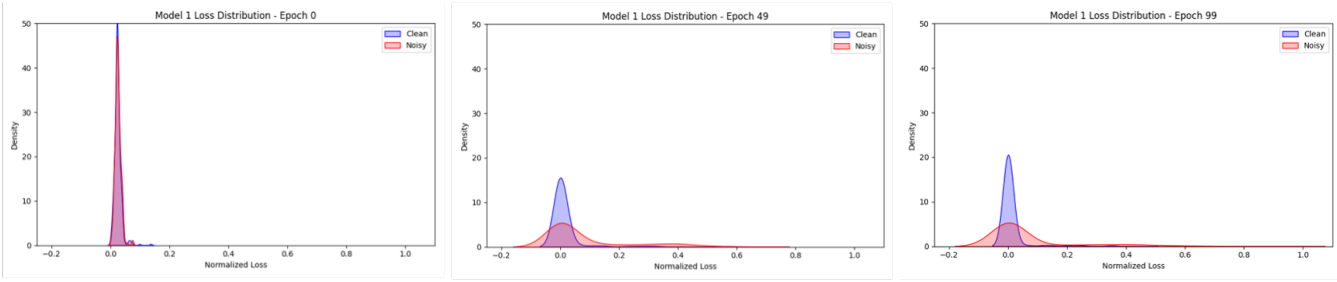
4.3.3 Loss Distributions of Clean vs. Noisy Samples. Figure 7 displays the loss distributions of clean and noisy samples under the 35% unstructured noise setting at three training stages (early, middle, and late epochs). As shown, the overlap between clean and noisy samples remains substantial throughout training, indicating that Co-teaching struggles to separate clean from mislabeled samples in this clinical dataset. To provide a broader perspective, additional loss distribution plots for 20%, 30%, and 45% noise settings are included in Supplementary Figure13, 14 and 15. These show similar trends: at low noise levels, the distributions are more distinct, allowing Co-teaching to filter noisy samples more effectively, whereas at higher noise levels, the overlap increases. These patterns were further supported by trends in the average training loss and pure ratio over epochs (see Supplementary Figures 23, 24, 25, 26 and 34), which reflect training instability and reduced label quality under severe noise conditions.

4.3.4 Comparison to OrganMNIST3D. While Co-teaching demonstrated robust performance on the OrganMNIST3D dataset under both symmetric and pairflip noise, its effectiveness on the ADNI dataset was notably more constrained. In ADNI, for instance, Co-teaching outperformed the baseline at 20% unstructured noise (77.43% vs. < 61.5%), but the gain decreased at higher noise levels and completely reversed at 40% structured noise. This pattern is in contrast with the results on the OrganMNIST3D dataset, where Co-teaching consistently outperformed the baseline up to 50% symmetric and 45% pairflip noise. Comparing the loss distribution plots of the two datasets revealed that Co-teaching was better able to eliminate mislabelled samples on OrganMNIST3D. In this dataset, clean and noisy samples were more easily separable by loss, allowing Co-teaching to discard mislabelled samples more effectively. In contrast, in the ADNI experiments, loss distributions showed substantial overlap between clean and noisy samples (see Figure 7), limiting the model's ability to identify label noise reliably. The pure ratios plots and the average training loss over epochs plots confirm this and suggest greater training instability and less confident filtering in ADNI compared to OrganMNIST3D.

4.4 ADNI: Structured Noise

4.4.1 Test Accuracy Across Noise Conditions. Co-teaching was also applied on the ADNI dataset under structured label noise (age-dependent flipping) at flipping rates of 10%, 20%, 30%, 35%, and 40%. Table 4 shows that Co-teaching outperformed the baseline at all noise levels except at 40%, with the largest improvement seen at 30% flipping rate: 77.88% vs. 66.37%. However, as shown in supplementary Table 7, at the 30% flipping rate, the final test accuracy and the average Accuracy over the last 10 epochs differ by nearly 10%, indicating some instability towards the end of training. Moreover, similar to the results observed for ADNI under unstructured noise and for OrganMNIST3D, the balanced Accuracy showed minimal differences of less than 2% compared to the overall test accuracy, suggesting that class imbalance did not significantly distort the evaluation.

At the lowest noise levels (10% and 20%), both models performed similarly (e.g., 80.53% vs. 72.57% at 10%), but Co-teaching maintained higher performance as noise increased up until 35%. At the highest



(a) (b) (c)

Figure 7: ***Loss distribution of Model 1 on ADNI with 35% unstructured noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 49, 99). There remains substantial overlap between clean and noisy losses.***

Table 3: Final test accuracy, balanced accuracy (in %), and AUC for ADNI under unstructured noise. All values are from the last training epoch.

Flipping Rate	Baseline			Co-teaching		
	Acc (%)	Balanced Acc (%)	AUC	Acc (%)	Balanced Acc (%)	AUC
20%	61.5 [54.4–67.3]	61.4 [54.9–67.6]	0.65 [0.56–0.72]	77.4 [71.2–82.3]	76.7 [71.4–81.7]	0.83 [0.77–0.88]
30%	59.7 [52.7–65.5]	59.6 [53.0–65.9]	0.63 [0.56–0.70]	63.3 [56.6–69.0]	63.3 [56.7–69.4]	0.62 [0.55–0.69]
35%	55.8 [48.7–61.9]	55.4 [48.9–61.7]	0.56 [0.48–0.63]	54.4 [47.3–60.2]	55.5 [49.7–61.0]	0.60 [0.52–0.67]
45%	43.4 [36.3–49.6]	43.1 [36.7–49.5]	0.43 [0.35–0.50]	56.6 [49.6–62.8]	55.3 [50.0–60.6]	0.60 [0.53–0.68]

noise level (40%), Co-teaching performs worse (55.31%) than the baseline (63.27%), suggesting that higher levels of structured noise may hinder the model's ability to select clean samples.

4.4.2 Statistical Significance of the Results. The confidence intervals in Table 4 indicate that at a 30% noise rate, Co-teaching achieves statistically significant improvement for all metrics except balanced accuracy. However, at noise levels of 10%, 20%, and 40%, the confidence intervals partially overlap, indicating that the differences are not statistically significant.

4.4.3 AUC performance. In 4, AUC scores for Co-teaching are higher than those of the baseline at all but the highest noise level, confirming its effectiveness under moderate structured noise conditions. These findings suggest that Co-teaching might offer robustness to structured label noise when applied to the ADNI dataset, but its performance may worsen under higher noise levels.

4.4.4 Loss Distributions of Clean vs. Noisy Samples. Figure 8 shows the loss distributions of clean versus noisy samples at early, middle, and late training epochs for the 30% structured noise setting. This specific condition was selected for visualization because it yielded the highest test accuracy improvement for Co-teaching over the baseline (77.88% vs. 66.37%) and was the only statistically significant case based on the confidence intervals. However, as discussed, in Table 7, the average Accuracy over the last 10 epochs was slightly lower for Co-teaching than for the baseline (67.83% vs. 68.19%), indicating some fluctuations in performance towards the end of training. This assumption is confirmed by the trends in the average training loss and pure ratio over epochs (see Supplementary Figures 27, 28, **??**, **??**, **??**, and 35). The loss distributions reveal partial separation between clean and noisy samples during training. While this separation is sufficient to yield performance improvements at moderate noise levels, substantial overlap remains throughout, especially in later epochs. This highlights the inherent limitation of loss-based filtering under structured noise conditions, where label ambiguity makes it difficult to identify sample with clean labels consistently. Additional loss distributions for other structured noise levels (10%, 20%, 35%, and 40%) are provided in the Supplementary Figure 16, 17, 18 and 19

4.4.5 Comparison to OrganMNIST3D and ADNI Unstructured. While the Co-teaching method achieved strong improvements at 30% structured noise (77.88% vs. 66.37%) on the ADNI dataset, its performance dropped sharply at 40%, where it underperformed the baseline. In contrast, on OrganMNIST3D, Co-teaching outperformed the baseline under all three tested noise conditions, suggesting a more stable trend in this setting. When comparing the ADNI structured noise results to ADNI under unstructured noise, Co-teaching appears more robust in the unstructured setting. For example, at 45% noise, it still maintained a notable performance advantage over the baseline (56.64% vs. 43.36%), whereas under 40% structured noise, the method performed worse than the baseline. This suggests that unstructured noise may be easier to filter using loss-based methods, while structured noise, especially when correlated with features like age, can mislead the model's assumptions about which samples are reliable. The loss distributions further support this difference. Under structured noise, clean and noisy samples show some early separation but substantial overlap toward the end of training, especially at higher noise levels (see Figure 8). In contrast, the 35% unstructured setting showed a clearer distinction between clean

and noisy samples over time (see Figure 7).However, the overlap remained substantial, reflecting the difficulty of separating ambiguous clinical labels regardless of the underlying noise structure.

5 DISCUSSION

The purpose of this study was to evaluate the effectiveness of Co-teaching in medical settings by identifying the conditions under which Co-teaching improves model performance in the presence of label noise. The goal was to determine when the method is most suitable for tackling diagnostic uncertainty in medical imaging, specifically in Alzheimer's Disease Dementia. Our findings suggest that the effectiveness of co-teaching strongly depends on the dataset properties and the type and severity of label noise.

5.1 Summary of Key Findings

Across all classification tasks, the Co-Teaching method showed improved performance over the baseline in most conditions, but especially under higher levels of unstructured label noise. This improvement was mostly shown in the OrganMNIST3D dataset, which features clear class boundaries and relatively clean labels. This is consistent with prior work, where standard models such as ResNet-18 and ResNet-50 achieved test accuracies up to 90.7% and AUCs as high as 0.996 on this dataset [28, 29]. However, under structured noise settings, Co-teaching tends to perform less reliably than under unstructured noise, especially in datasets with ambiguous class boundaries. Performance gains were more modest in the ADNI dataset, probably due to inherent diagnostic ambiguity in clinical data. Co-Teaching occasionally performed worse than the baseline on the AD vs. CN classification task. Co-teaching removes high-loss samples under the assumption that they are mislabeled. However, in clinical datasets, such samples can still be correctly labeled and carry meaningful information due to diagnostic uncertainty. The baseline model, on the other hand, retains all samples, including those discarded by Co-teaching. Its performance advantage may stem from correctly labeled samples that Co-teaching prematurely excluded. Moreover, in the OrganMNIST3D dataset, the loss distributions for noisy and clean samples were more clearly separable, aligning with the observed performance gains from Co-Teaching. In contrast, in the ADNI task, there was substantial overlap between the loss distributions of clean and noisy samples, suggesting that loss is a less reliable proxy for label correctness in complex clinical domains. The findings also indicate that Co-Teaching is most effective when clean samples consistently produce lower losses, an assumption that does not always hold in datasets like ADNI. These findings suggest that Co-Teaching is most effective under specific conditions. Below, we provide practical guidelines derived from our results to help determine when Co-Teaching is likely to be beneficial:

- **Use in datasets with clearly defined class boundaries and low ambiguity:**
  When used in in settings with clearly separated class features and low ambiguity. Co-Teaching works best when features of each class are distinct and well-separated , such as in the OrganMNIST3D dataset. This method works especially well under high levels of symmetric noise. Therefore, this method could be suited for scenarios where noise is introduced through manual data entry in clinical registries. In these cases, mislabeled samples are more likely to be true outliers and can be reliably identified based on their training loss.
- **Best suited for moderate levels of structured noise**
  Co-Teaching appears to perform best when the level of structured label noise is moderate. In such cases, there is enough noise for the method to make a difference, but not so much that training becomes unstable or valuable clean samples are discarded. This balance allows Co-Teaching to effectively filter mislabeled data without significantly harming performance. It may also be suitable for scenarios where labels are systematically but inconsistently misassigned, such as clinical coding errors. In these cases, the structured nature of the noise still allows the method to distinguish between clean and noisy samples, provided the overall noise level remains moderate.

5.2 Limitations of Co-Teaching

Despite its strengths, Co-Teaching demonstrates important limitations when applied to clinical datasets. The following sections discuss key factors that limit its effectiveness.

5.2.1 Discarding high-loss samples too early. Co-teaching occasionally performed worse than the baseline on the AD vs. CN classification task. This could be because Co-teaching eliminates high-loss samples early in training ,some of which are in fact correctly labelled. In clinical datasets with inherent diagnostic uncertainty, such samples may produce high loss not because they are mislabelled, but because they are difficult to learn. While Co-teaching discards these samples, the baseline model retains them and thereby benefits from clean examples that Co-teaching mistakenly removes.

5.2.2 Loss as an unreliable proxy in clinical domains. Moreover, analysis of loss distributions showed that, in the ADNI tasks, there was limited separation between clean and noisy samples. This suggests that loss is a less reliable proxy for label correctness in complex clinical domains.

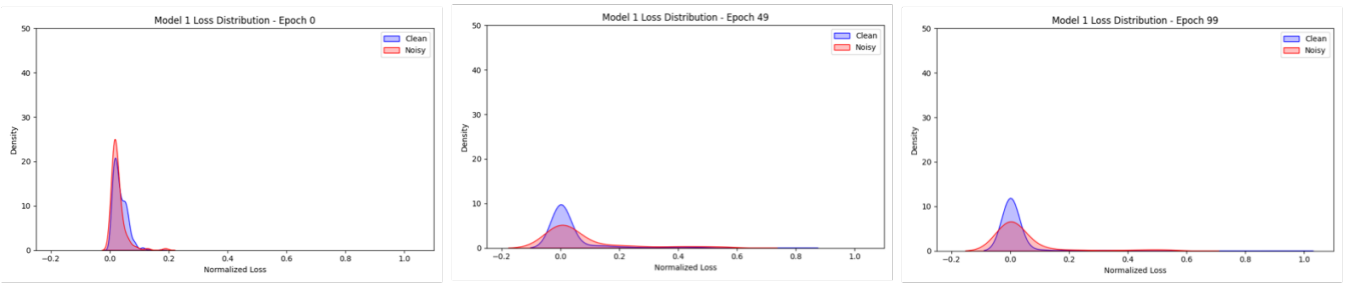
5.2.3 Dependency on dataset characteristics. The findings also indicate that Co-Teaching is most effective when clean samples consistently produce lower losses, an assumption that does not always hold in datasets like ADNI. Since the effectiveness of Co-Teaching depends heavily on the context, such as the simplicity of the task and the nature and amount of label noise, it cannot be reliably applied to new datasets where these properties are unknown or hard to estimate. This makes it less useful in practical applications, especially when working with complicated and varied datasets like ADNI.

5.3 Challenges in Ground Truth Quality and Evaluation Metrics

Beyond the limitations of the Co-Teaching method itself, this study also faced challenges concerning both the quality of the ADNI labels and the interpretability of the pure ratio as an evaluation metric. Although ADNI was designed as a large-scale, observational study focused on biomarker discovery and disease monitoring, its

Table 4: Final test accuracy, balanced accuracy (in %), and AUC for ADNI under structured noise. Actual noise rates are reported in parentheses. Confidence intervals (95%) are reported next to the corresponding metric values.

Flipping Rate (Actual)	Baseline			Co-teaching		
	Acc (%)	Balanced Acc (%)	AUC	Acc (%)	Balanced Acc (%)	AUC
10% (4.81%)	72.6 [65.9–77.4]	71.9 [66.3–77.2]	0.82 [0.763–0.868]	80.5 [74.8–85.0]	80.5 [75.2–85.4]	0.86 [0.81–0.90]
20% (9.62%)	72.6 [65.9–77.4]	72.6 [66.7–78.3]	0.80 [0.734–0.852]	76.1 [69.9–81.0]	75.9 [70.2–81.4]	0.85 [0.79–0.89]
30% (14.42%)	66.4 [59.7–71.7]	66.9 [60.8–72.7]	0.79 [0.718–0.841]	77.9 [71.7–82.3]	77.4 [72.0–82.3]	0.82 [0.76–0.87]
35% (16.73%)	60.2 [53.1–65.9]	60.8 [54.8–66.9]	0.65 [0.575–0.716]	65.0 [58.0–70.8]	66.2 [61.2–71.3]	0.83 [0.77–0.88]
40% (19.23%)	63.3 [56.6–69.0]	63.8 [57.6–69.6]	0.72 [0.645–0.777]	55.3 [48.2–61.1]	56.9 [52.2–61.5]	0.72 [0.65–0.75]



(a) (b) (c)

Figure 8: ***Loss distribution of Model 1 on ADNI with 30% structured noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 49, 99). Loss distributions remain overlapped even at later epochs.***

diagnostic labels, such as CN, MCI, or AD, were in most cases not pathologically confirmed, with only a small subset of cases having confirmation through autopsy. As Potter et al. (2024) note, ADNI classifications were mostly based on clinical and neuropsychological assessments, imaging data, and CSF or blood biomarkers This introduces inherent uncertainty into the labels, even before injecting noise. When noise is injected by flipping labels, the actual correctness of those flips cannot be determined. In some cases, a flipped label may better reflect the subject's true condition; In other cases, a previously correct label may be flipped into an incorrect one, thereby introducing new noise into the data. Thus, the effective noise rate may be either slightly higher or lower than the nominal flipping percentage. However, given that only a limited number of labels are impacted, this deviation is unlikely to significantly influence the overall analysis or the observed performance patterns.

Another challenge particularly present in the structured noise setting in ADNI is the validity of the pure ratio. When structured noise based on age is injected, the total number of incorrect labels in the dataset will decrease, since only a subsection of the dataset is affected by the noise. Although the actual noise rate was calculated and displayed in the results section, the pure ratio may still give a misleading impression of model performance, as it naturally increases when fewer noisy samples are present. This makes it an unreliable metric for comparing results across noise types or injection strategies, despite Han et al. (2018) using that metric as evidence for Co-Teaching performance.

5.4 Alternative Methods for Learning with Noisy Labels

The results of this study highlighted the need for alternative methods for Learning with Noisy Labels that bypass the limitations that the co-teaching method has. To address these shortcomings, recent research has proposed alternative strategies that either (1) retain all samples and use softer selection mechanisms, or (2) avoid relying on the loss function altogether by leveraging uncertainty, prediction consistency, or explainability. Below, we outline representative methods from both categories.

5.4.1 Retaining Noisy Samples with Smarter Selection. One significant disadvantage of Co-Teaching is its reliance on removing high-loss samples early in training. Although this method can effectively delete mislabeled data in environments with clear loss separation, it becomes problematic when clean and noisy samples produce comparable losses. In datasets like the ADNI, where diagnostic ambiguity and class overlap are widespread, this can result in the exclusion of informative but ambiguous samples. A method that addresses this issue is DivideMix. Unlike Co-Teaching, which removes samples based on their loss, DivideMix takes a softer approach by keeping all data and handling uncertainty more carefully. This is especially useful in situations where it is hard to clearly separate clean from noisy labels, something that is common in clinical datasets. Moreover, DivideMix introduces consistency regularization and label sharpening to further mitigate the impact of noise [12]. Because it adapts to the structure of the data during training, it is a particularly suitable method for challenging tasks such as Alzheimer's classification, where diagnostic uncertainty is common and the boundaries between classes are not always well

defined. Thus, DivideMix avoids premature rejection of valuable training data and offers a more adaptable and reliable framework for learning with noisy labels.

5.4.2 Beyond Loss: Explainable and Uncertainty-Based Methods. In Sorino et. al [22], a method is proposed using a trained model to classify patient data into three groups: Cognitively Normal (CN), Mild Cognitive Impairment (MCI), or Alzheimer's Disease (AD). The method focusses on the accuracy of the model and tries to interpret why the model arrived at a certain diagnosis by using explainable AI. The main technique used is SHAP (SHapley Additive exPlanations), a model-agnostic technique that quantifies the contribution of each feature to a specific prediction. SHAP values assign an importance score to every feature for each individual prediction, helping to interpret why the model arrived at a certain diagnosis. Rather than focusing solely on prediction accuracy, the method assesses how SHAP values develop over time throughout numerous clinical visits. It calculates the first-order ($\Delta 10$) and second-order ($\Delta 21$) differences in SHAP values between consecutive visits. These distinctions describe how the model's reasoning changes between time points, allowing the authors to distinguish between three sorts of transitions.

- **Stable trajectories:** Small and consistent SHAP variations across visits, suggesting cognitive status remained the same.
- **True cognitive changes:** Large shifts in SHAP values, reflecting genuine progression or improvement in cognitive function.
- **Potential label noise:** Cases where the diagnosis changes, but SHAP values remain stable — indicating the label may be incorrect or inconsistent.

Another method that does not rely on the loss function is Early-Learning Regularization (ELR). Although Co-Teaching has outperformed regularization-based methods such as ELR in high-noise benchmark datasets (e.g., CIFAR with symmetric noise) [24],it may not be the best option in real-world clinical contexts. In datasets like ADNI, where label noise is vague and class boundaries are unclear, ELR's reliance on early-prediction consistency rather than loss-based selection may be a more robust option. Unlike Co-Teaching, ELR does not discard samples and may be more appropriate for maintaining potentially informative, high-uncertainty cases [13].

6 FUTURE WORK

While this study focused on loss-based filtering through Co-teaching, future research should investigate alternative approaches for dealing with noisy labels in medical datasets. For instance, a comparative analysis with both hybrid approaches, such as DivideMix, and before- or non-loss-based approaches, like SHAP, could provide insight into whether the reliance on loss as a proxy for label correctness is valid in clinical datasets. Furthermore, future work should investigate alternative clinically grounded noise scenarios. For example, a focus on instance-dependent noise that simulates underdiagnosis in a younger population, where mild or early-stage cases are more likely to be mislabeled as healthy. Another promising direction is to introduce instance-dependent noise based on patient- or image-specific features, such as age, comorbidities, or subtle anatomical patterns, reflecting how diagnostic difficulty may vary across subpopulations.

7 CONCLUSION

The aim of this research was to investigate the effectiveness of the Co-teaching method when it is applied to noisy medical imaging datasets. This study showed that Co-Teaching can significantly improve performance under both structured and unstructured label noise, particularly in settings where class boundaries correspond to clear anatomical differences in the images, as is the case in organ classification, and when label noise is synthetic. For example, the multi-class organ classification task (OrganMNIST3D) tends to provide more favorable conditions for noise-robust methods like Co-Teaching. This is possibly due to the dataset's visually distinct anatomical categories and relatively low diagnostic ambiguity. However, in more complex settings, such as Alzheimer's disease classification using the ADNI dataset, the overlapping features and diagnostic uncertainty make it more difficult to separate clean from noisy labels effectively. While Co-Teaching can offer benefits in the presence of label noise, its effectiveness is highly context-dependent. In particular in tasks with inherent diagnostic uncertainty, such as dementia prediction, per-sample loss may not be a reliable proxy for label correctness. Unlike well-separated tasks like organ classification, ambiguous or borderline cases in dementia tend to exhibit high loss even when labels are not necessarily incorrect. This limits the ability of Co-teaching to identify truly noisy samples, and may even lead to the exclusion of valuable training examples. In summary, Co-teaching can be effectively applied to medical imaging tasks, including more complex ones such as Alzheimer's disease classification, under the right conditions. It works best on tasks with low inter-class ambiguity and relatively simple label noise, such as symmetric noise, where the source and pattern of the noise are easier to model and account for. Even in clinically challenging domains, such as ADNI, Co-teaching showed improvements under moderate noise levels and in structured noise settings. However, in settings where clean and noisy samples are less distinguishable, particularly due to diagnostic uncertainty, loss may become a less reliable proxy for sample correctness.

8 ACKNOWLEDGMENTS

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9 AVAILABILITY

All datasets used in this study are publicly available and can be accessed through the following websites :

- CIFAR-10: https://www.cs.toronto.edu/ kriz/cifar.html
- OrganMNIST3D: Available via MedMNIST v2: https://medmnist.com/
- ADNI: Accessible upon request from the Alzheimer's Disease Neuroimaging Initiative: https://adni.loni.usc.edu/

The code for this project will be made available at: www.gitlab.com/moetnogkomen

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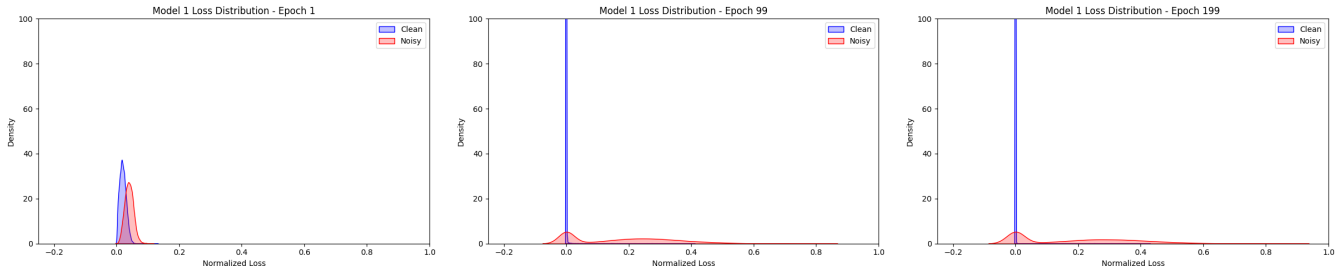
10 SUPPLEMENTAL INFORMATION

Table 5: Test accuracy on OrganMNIST3D under different noise types. Reported are the average of the last 10 epochs and their standard deviation.

Noise type	Baseline_OrganMNIST3D	Co-teaching_OrganMNIST3I
Pairflip 45%	52.72% (51.18% ± 1.57%)	57.31% (57.08% ± 2.28%)
Symmetric 50%	64.18% (65.10% ± 0.61%)	75.36% (74.96% ± 1.78%)
Symmetric 20%	89.68% (89.97% ± 0.60%)	89.97% (89.00% ± 2.17%)

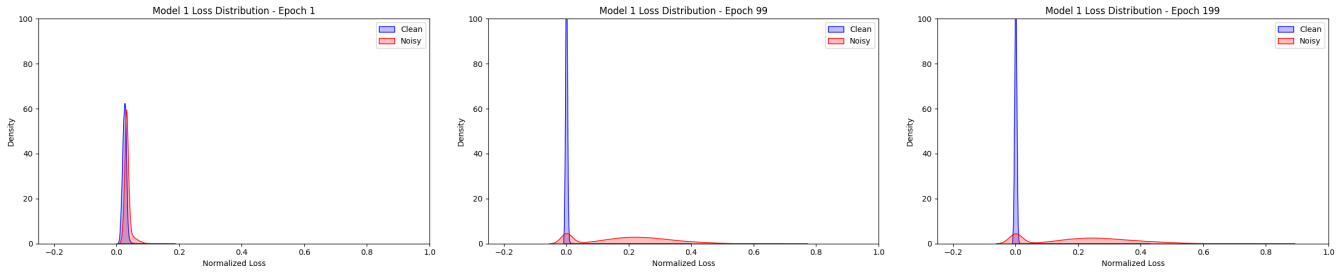
Table 6: Test accuracy on ADNI with unstructured noise. Reported are the average of the last 10 epochs and their standard deviation.

Flipping rate	Baseline_ADNI	Co-teaching_ADNI
45%	43.36% (48.19% ± 2.34%)	56.64% (54.95% ± 1.08%)
35%	55.75% (55.31% ± 4.75%)	54.42% (57.87% ± 2.85%)
30%	59.73% (60.53% ± 0.59%)	63.27% (56.68% ± 3.17%)
20%	61.50% (58.54% ± 2.70%)	77.43% (76.24% ± 1.53%)



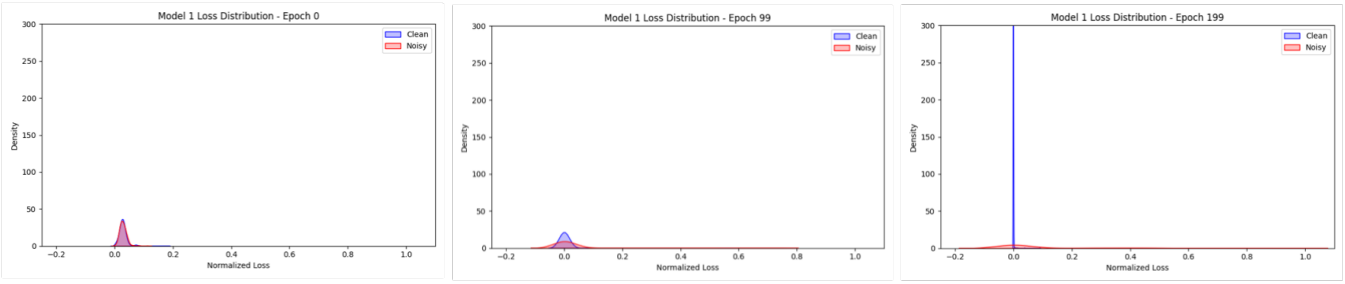
(a) (b) (c)

Figure 9: Loss distribution of Model 1 on CIFAR-10 with 20% symmetric noise. Comparison of clean vs noisy samples at three training stages (Epoch 1, 99, 199).



(a) (b) (c)

Figure 10: Loss distribution of Model 1 on CIFAR-10 with 50% symmetric noise. Comparison of clean vs noisy samples at three training stages (Epoch 1, 99, 199).

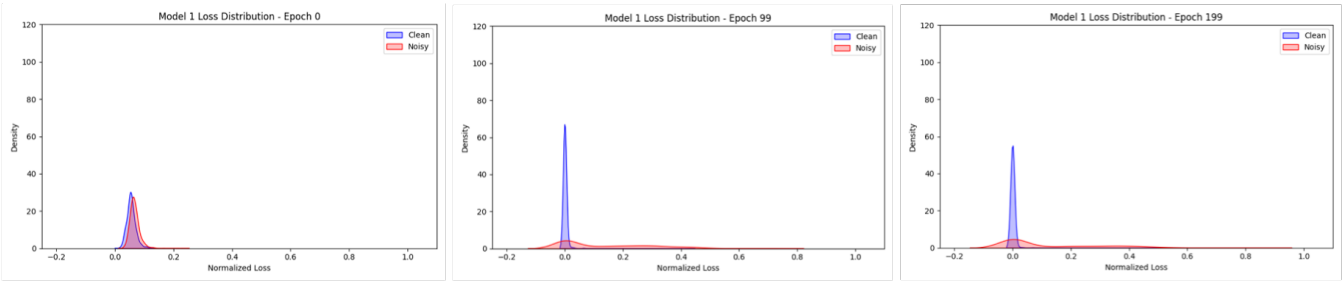


(a) (b) (c)

Figure 11: Loss distribution of Model 1 on OrganMNIST3D with 20% symmetric noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 99, 199).

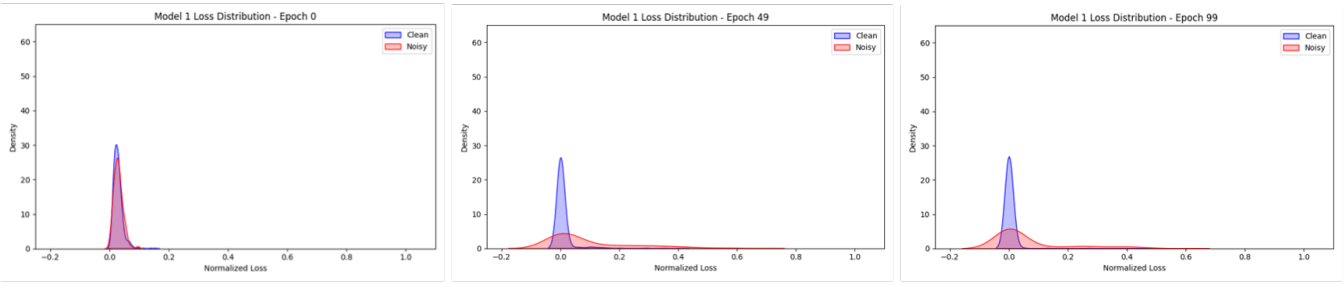
Table 7: Test accuracy on ADNI with structured (instance-dependent) noise. Reported are the average of the last 10 epochs and their standard deviation.

Flipping rate (actual)	Baseline_ADNI	Co-teaching_ADNI
40% (19.23%)	63.27% (64.51% ± 2.72%)	55.31% (58.41% ± 6.87%)
35% (16.73%)	60.18% (58.94% ± 1.92%)	65.04% (68.45% ± 3.45%)
30% (14.42%)	66.37% (68.19% ± 1.14%)	77.88% (67.83% ± 4.12%)
20% (9.62%)	72.57% (70.58% ± 3.55%)	76.11% (71.51% ± 4.48%)
10% (4.81%)	72.57% (74.07% ± 3.36%)	80.53% (76.46% ± 4.27%)



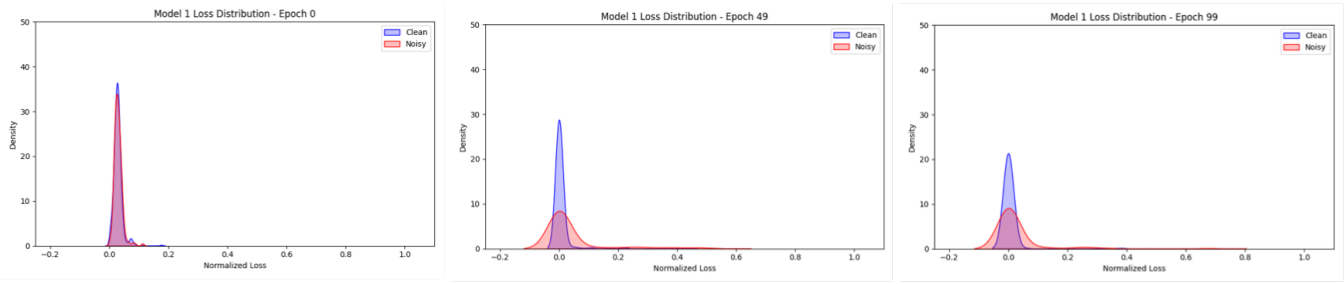
(a) (b) (c)

Figure 12: Loss distribution of Model 1 on OrganMNIST3D with 50% symmetric noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 99, 199).



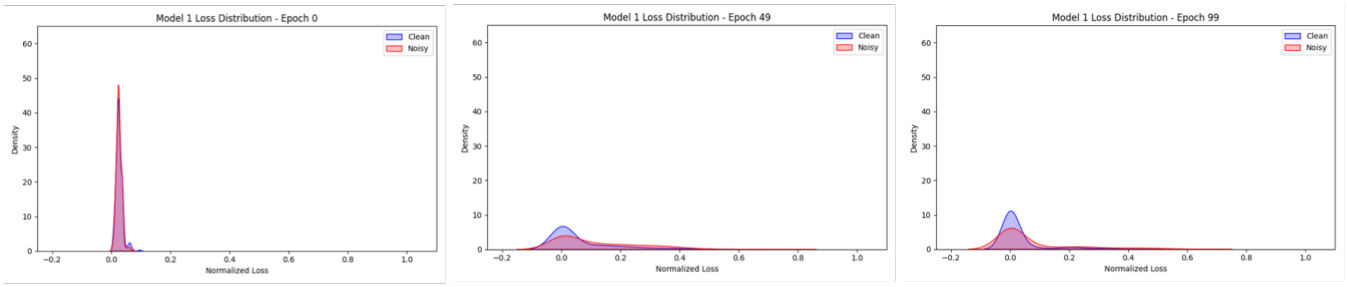
(a) (b) (c)

Figure 13: Loss distribution of Model 1 on ADNI under 20% unstructured noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 49, 99).



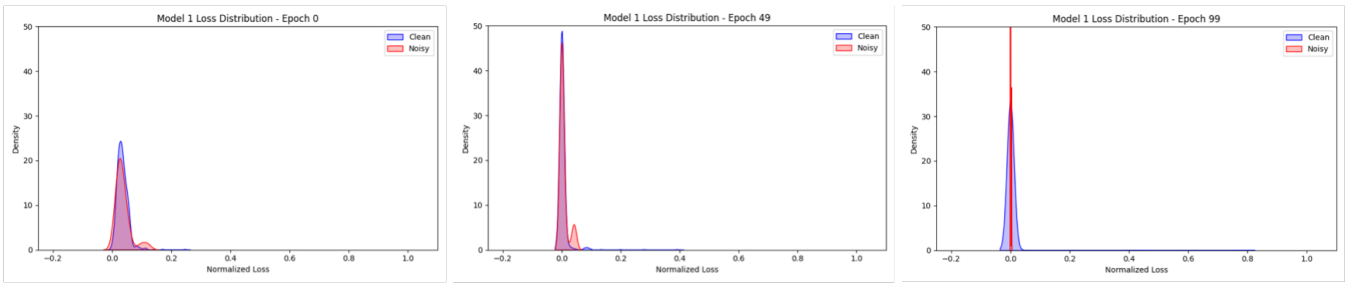
(a) (b) (c)

Figure 14: Loss distribution of Model 1 on ADNI under 30% unstructured noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 49, 99).



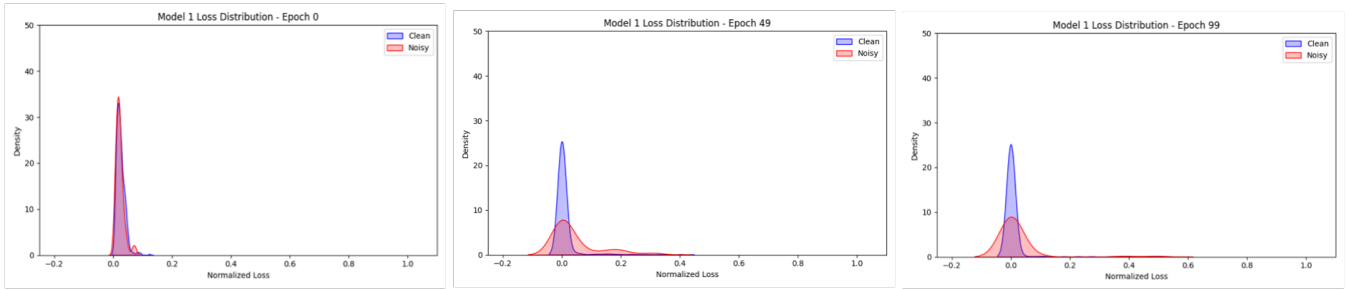
(a) (b) (c)

Figure 15: Loss distribution of Model 1 on ADNI under 45% unstructured noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 49, 99).



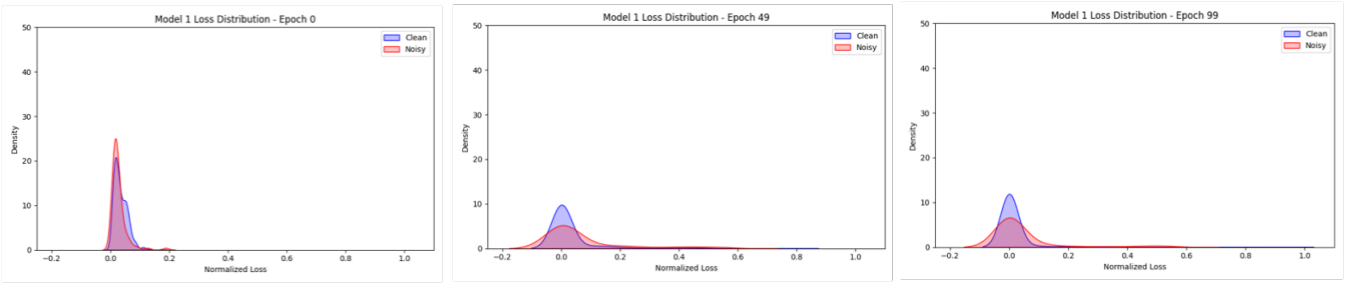
(a) (b) (c)

Figure 16: Loss distribution of Model 1 on ADNI under 10% structured noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 49, 99).



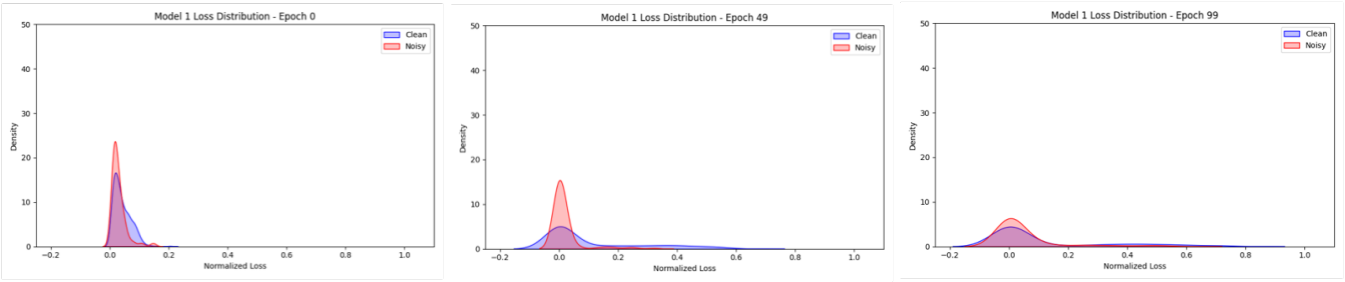
(a) (b) (c)

Figure 17: Loss distribution of Model 1 on ADNI under 20% structured noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 49, 99).



(a) (b) (c)

Figure 18: Loss distribution of Model 1 on ADNI under 35% structured noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 49, 99).



(a) (b) (c)

Figure 19: Loss distribution of Model 1 on ADNI under 40% structured noise. Comparison of clean vs noisy samples at three training stages (Epoch 0, 49, 99).

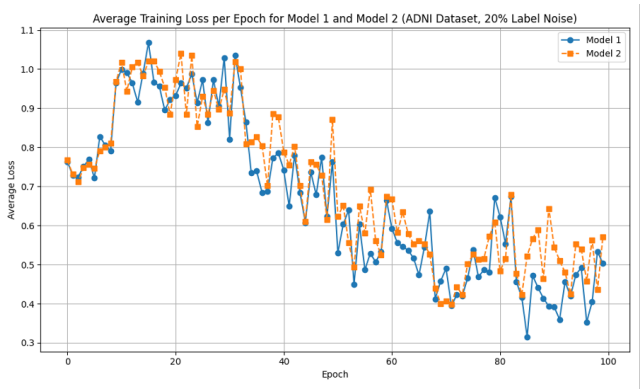


Figure 23: Average training loss per epoch for Model 1 and Model 2 on the ADNI dataset under 20% unstructured noise.

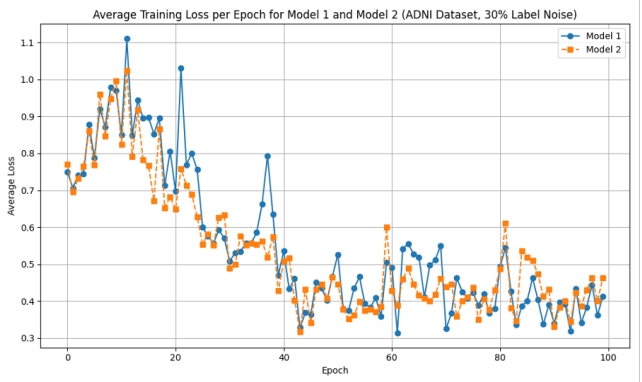


Figure 24: Average training loss per epoch for Model 1 and Model 2 on the ADNI dataset under 30% unstructured noise.

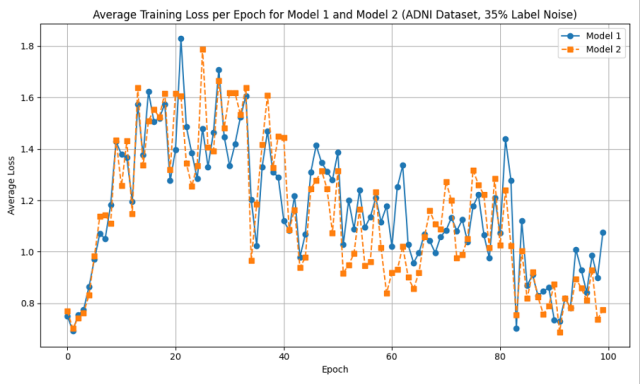


Figure 25: Average training loss per epoch for Model 1 and Model 2 on the ADNI dataset under 35% unstructured noise.

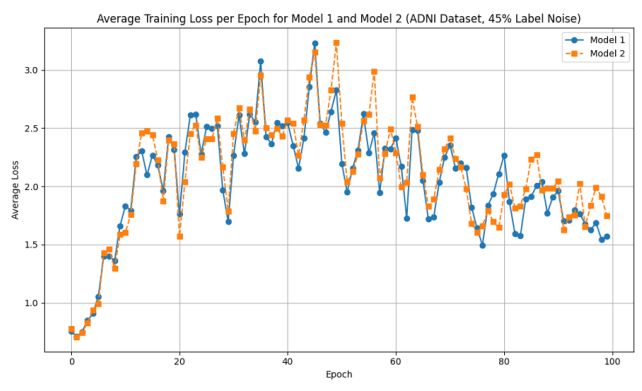


Figure 26: Average training loss per epoch for Model 1 and Model 2 on the ADNI dataset under 45% unstructured noise.

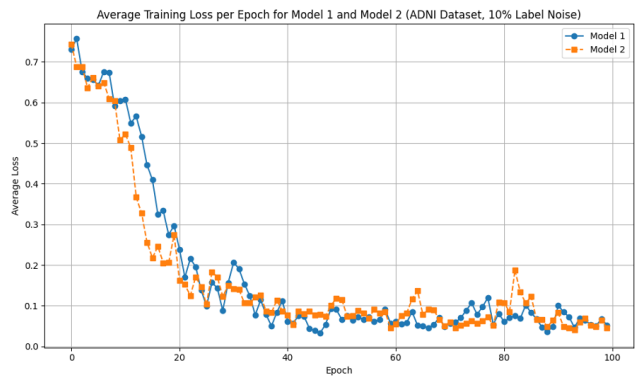


Figure 27: Average training loss per epoch for Model 1 and Model 2 on the ADNI dataset under 10% structured noise.

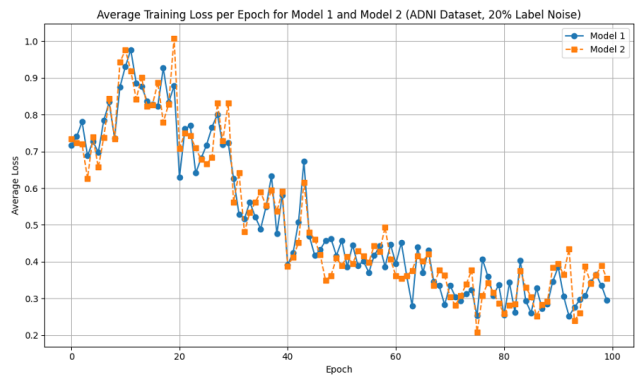


Figure 28: Average training loss per epoch for Model 1 and Model 2 on the ADNI dataset under 20% structured noise.

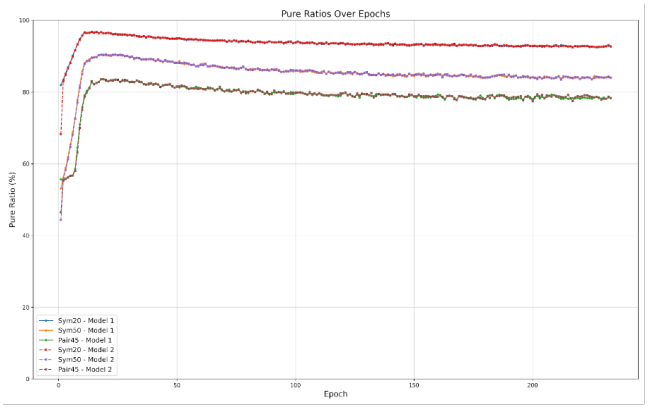


Figure 32: Pure ratios over epochs for CIFAR-10 under different noise types and levels. Symmetric noise (especially 20%) leads to higher and more stable pure ratios, while pairflip noise results in lower pure ratios.

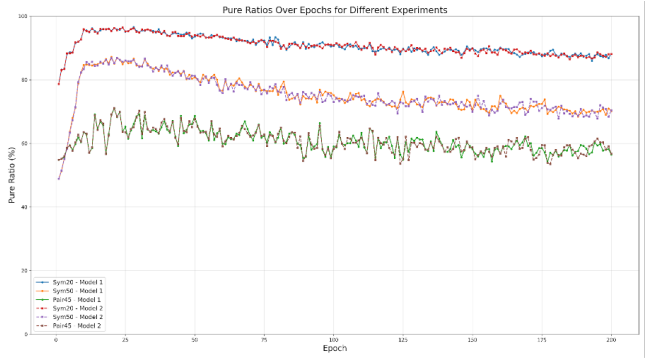


Figure 33: Pure ratios over epochs for OrganMNIST3D under various noise conditions. Pairflip noise results in lower and noisier pure ratios compared to symmetric noise.

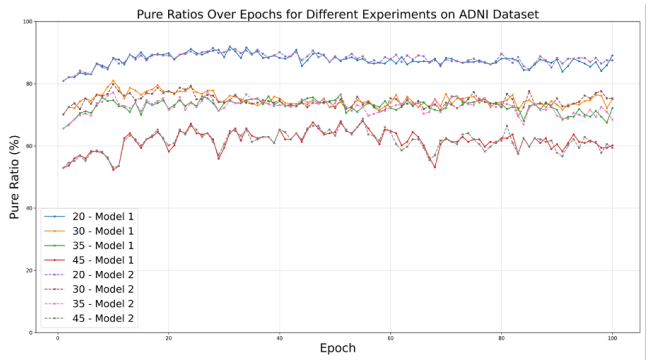


Figure 34: Pure ratios across different unstructured noise levels on the ADNI dataset. As noise increases, pure ratios decline and fluctuate more.

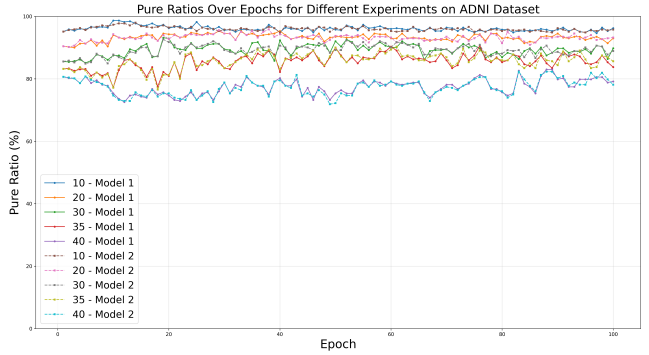


Figure 35: Pure ratio over epochs for the ADNI dataset with structured (instance-dependent) label noise. Higher noise levels lead to less stable pure ratios.